

GLOBAL EMISSIONS AND MODELS OF PHOTOCHEMICALLY ACTIVE COMPOUNDS

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ABSTRACT

Anthropogenic emissions from industrial activity, fossil fuel combustion, and biomass burning are now known to be large enough, relative to natural sources, to perturb the chemistry of vast regions of the troposphere. A goal of the IGAC Global Emissions Inventory Activity (GEIA) is to provide authoritative and reliable emissions inventories on a $1^\circ \times 1^\circ$ grid. When combined with atmospheric photochemical models, these high quality emissions inventories may be used to predict the concentrations of major photochemical products. Then, comparison of model results with measurements of pertinent species will allow us to assess whether there are major shortcomings in our understanding of tropospheric photochemistry, the budgets and transport of trace species, and their effects in the atmosphere. Through this activity, we are building the capability to make confident predictions of the future consequences of anthropogenic emissions. This paper compares IGAC recommended emissions inventories for reactive nitrogen and sulfur dioxide to those that have been in use previously. We also present results from the three-dimensional LLNL atmospheric chemistry model that show how emissions of anthropogenic nitrogen oxides might potentially affect tropospheric ozone and OH concentrations, and how emissions of anthropogenic sulfur increase sulfate aerosol loadings.

INTRODUCTION

Two issues have received considerable attention with regard to the potential role of changing anthropogenic emissions to global tropospheric photochemistry. The first issue concerns the oxidizing capacity of the atmosphere and our potential to change this capacity

through emissions of methane (CH_4), nitrogen oxides (NO_x), carbon monoxide (CO), and non-methane hydrocarbons (NMHCs). The oxidizing capacity of the atmosphere is determined mainly by the concentration of hydroxyl radicals (OH), although ozone (O_3) and hydrogen peroxide (H_2O_2) also play important roles. The hydroxyl radical reacts with a suite of species (Table 1), initiating photochemical reaction sequences that eventually lead to non-reactive products (e.g., CO_2) or to products that are removed by precipitation and dry deposition. If the OH concentration is decreased globally, the concentrations of gases removed by reaction with OH may increase even with no change in their sources. Hydroxyl radical formation is initiated by photolysis (at wavelengths shorter than about 310 nm) of O_3 which produces $\text{O}(^1\text{D})$. $\text{O}(^1\text{D})$ reacts with H_2O to form two hydroxyl radicals. Since the concentration of OH in the background atmosphere is determined mainly by the concentrations of CO , CH_4 , NO_x , and H_2O , it is important to quantify the anthropogenic and natural emissions of these species to determine their effects on O_3 and OH . Anthropogenic emissions of each are tied to industrialization. Increases in NO_x emissions are particularly important in changing oxidant concentrations because the local concentration of NO_x determines whether photochemical sequences lead to net O_3/OH increases or decreases (Penner, 1989; Crutzen and Zimmerman, 1991). Besides its role as an oxidant, changes to tropospheric ozone are of interest because of its role in OH chemistry and because it acts as a greenhouse gas and has deleterious health and ecosystem effects (e.g., Fishman *et al.*, 1979; Tilton, 1989).

Table 1. Species which are removed or oxidized by reaction with OH . DMS is $(\text{CH}_3)_2\text{S}$, CCN is cloud condensation nucleus, NMHC is non-methane hydrocarbon, HCFCs are hydrochlorofluorocarbons, and CFCs are chlorofluorocarbons.

Species	Importance
CH_4	Greenhouse gas; Affects tropospheric O_3 and OH ; Affects stratospheric O_3
CO	Urban pollutant; Affects tropospheric O_3 and OH cycles
CH_3CCl_3 , CH_3Br , CH_3Cl	Greenhouse gases; Release chlorine or bromine in stratosphere which destroys O_3
HCFCs	Greenhouse gases; Replacements for CFCs which release chlorine in the stratosphere and destroy O_3
NMHCs	Urban pollutants; Enhance tropospheric O_3 in high NO_x areas; Sink for tropospheric OH
DMS, H_2S , SO_2	Form sulfate aerosols which contribute to acid rain and reflect solar radiation; Form CCN which alter cloud properties and cool the climate

The second major issue for tropospheric chemistry, which has emerged from a consideration of changing anthropogenic emissions, concerns the formation of anthropogenic aerosols and their potential to mask part of the warming expected from greenhouse gas emissions. Anthropogenic aerosols are composed of a variety of aerosol types (e.g., water-soluble inorganic species, organic condensed species, elemental or black carbon, and mineral dust), but the best-quantified and probably the largest component of the anthropogenic aerosol is that derived through photochemical interactions of anthropogenic SO_2 emissions (Charlson *et al.*, 1992). Global average climate forcing through direct reflection of solar radiation by anthropogenic sulfate aerosols may range from -0.3 to -1.1 Wm^{-2} (Penner *et al.*, 1993b). This forcing varies regionally and acts locally to mask anthropogenic greenhouse gas forcing. The magnitude of the calculated forcing varies depending on the underlying surface albedo, so it is important to estimate correctly the rate of the photochemical processes which convert SO_2 to SO_4^{2-} as the